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OPTIMISING AIRBNB PRICING STRATEGIES IN NEW YORK CITY:

A REGRESSION AND CLUSTERING ANALYSIS OF LISTING CHARACTERISTICS AND MARKET SEGMENTATION

Track 1: Classical Machine Learning (Regression and Clustering)

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Executive Summary

This report analyses Airbnb listing data for New York City to examine factors influencing pricing and identify actionable market segments within the platform. Pricing dynamics are shaped by multiple structural and behavioural factors within the marketplace (Ding, Niu and Choo, 2023). Using the AB_NYC_2019 dataset containing 48,884 cleaned listings, the study combines exploratory data analysis, regression modelling and clustering techniques. Results show that pricing is primarily driven by location and room type, with Manhattan and entire-home listings forming the premium tier. A multivariate linear regression model explains approximately 53 percent of price variation on the log scale, confirming that pricing follows identifiable patterns while still reflecting local complexity. K-Means clustering identifies four listing segments including low activity listings, standard listings, investment listings and high demand value listings. These findings suggest that Airbnb pricing should be segment-aware and location-sensitive, supporting host performance and platform competitiveness (Airbnb, 2023).

Business Question

How can Airbnb optimise pricing strategies across New York City based on location, property characteristics and availability patterns, and what actionable listing segments can be identified using clustering techniques?

Dataset and Methodology

The analysis used the AB_NYC_2019 dataset sourced from Kaggle containing New York City Airbnb listings (Cox, 2019). After removing zero-price errors and cleaning missing fields, the working dataset contained 48,884 listings. Missing values in reviews_per_month were filled with zero where listings had not yet received reviews. The last_review field was converted to a numerical variable representing days_since_review, and minimum_nights values were capped at 365 to avoid unrealistic outliers.

Exploratory data analysis revealed a heavily right-skewed price distribution, with a median price of $106 and mean of $153 as shown in Figure 1.

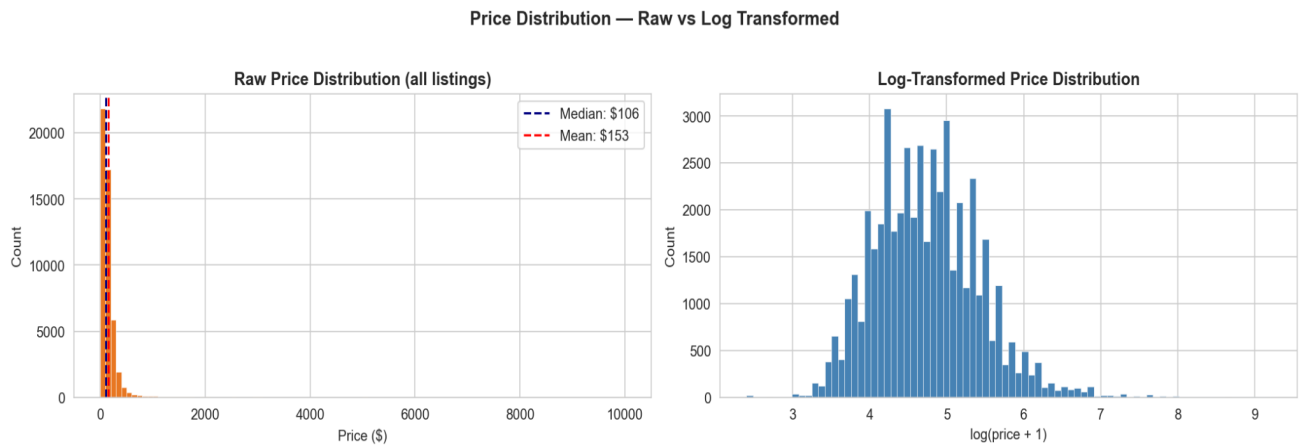

Figure 1. NYC Airbnb prices show strong right skew, supporting log transformation before modelling.

Exploratory analysis also revealed strong structural differences across boroughs and room types, as demonstrated in Figure 2. Manhattan listings consistently occupy the highest price range, while private and shared rooms form lower pricing tiers. These patterns indicate that location and accommodation type are fundamental drivers of pricing behaviour within the platform marketplace.

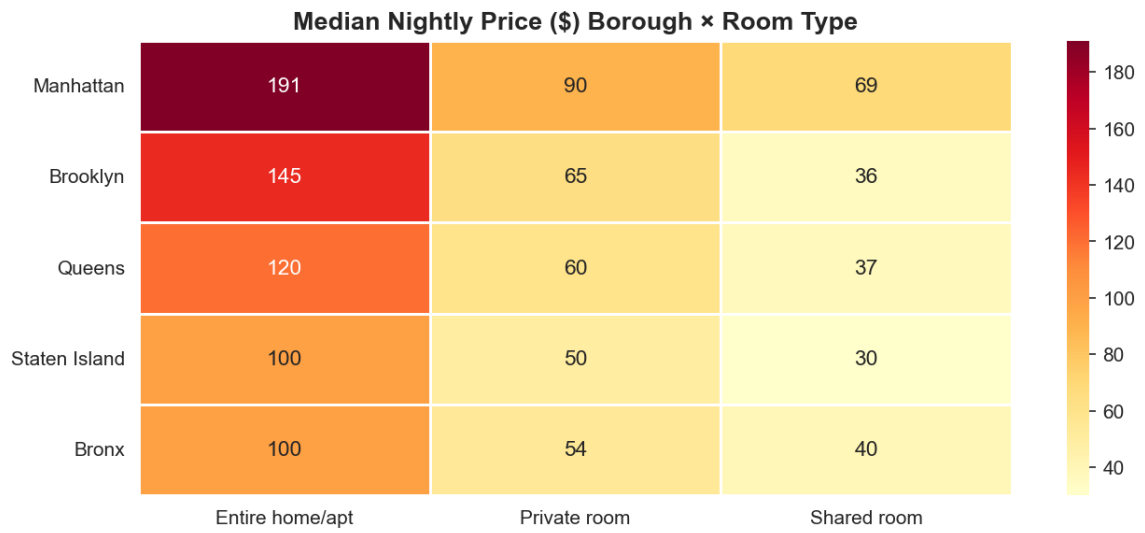

Figure 2. Median nightly prices vary by borough and room type, with Manhattan entire homes forming the premium tier.

Because regression models perform better with more normally distributed data, the price variable was log transformed before modelling. The modelling feature set included borough, neighbourhood, latitude, longitude, room type, minimum nights, number of reviews, review frequency, host portfolio size and availability indicators. Spatial patterns in pricing are illustrated in Figure 3, which highlights price hotspots concentrated in Manhattan and parts of Brooklyn.

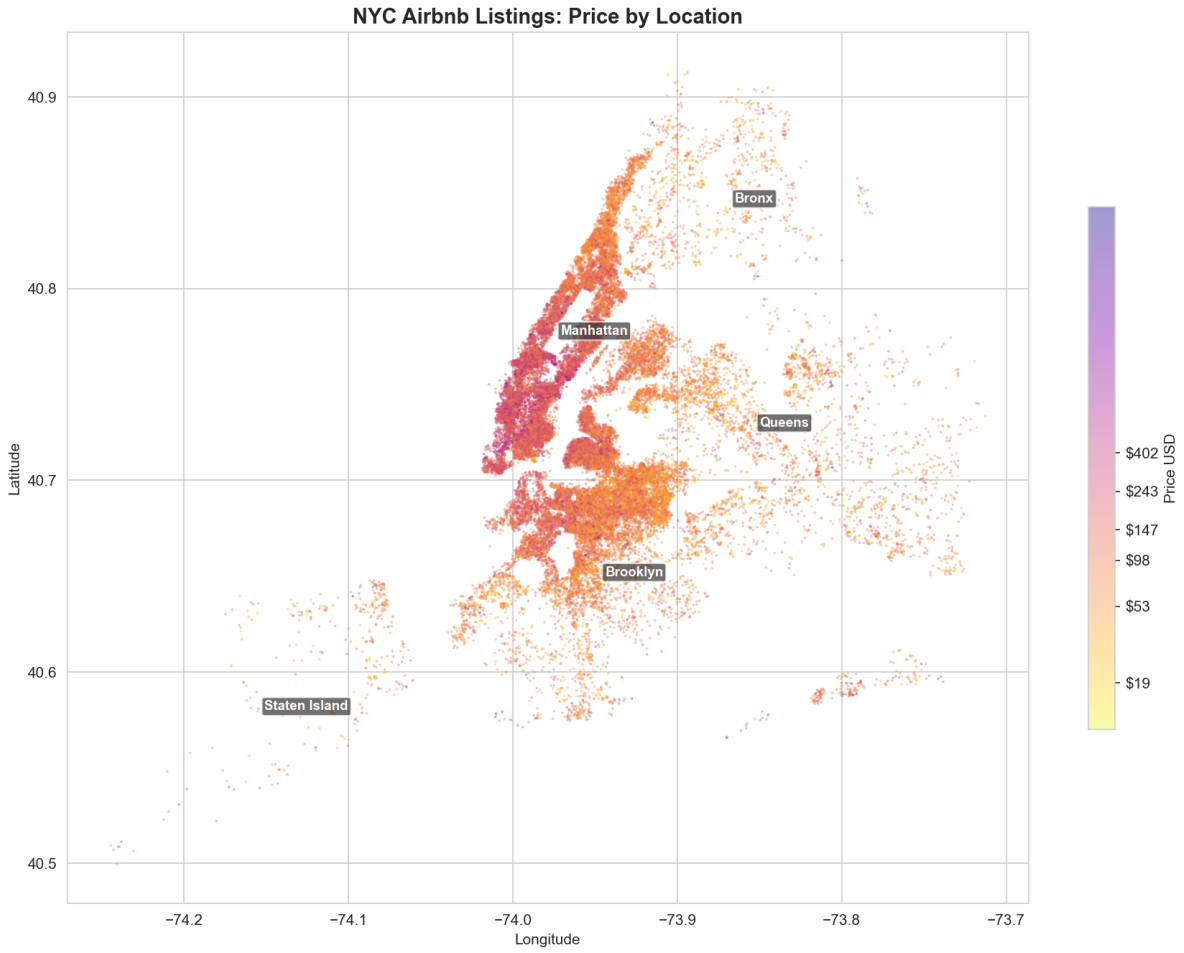

Figure 3. Geographic price hotspots concentrate in Manhattan and selected areas of Brooklyn.

Two classical machine learning approaches were applied. Linear regression was used to model price prediction because it provides transparent and interpretable results.

(James et al., 2021). K-Means clustering was used to identify natural groups of listings within the marketplace allowing the analysis to uncover different pricing strategies across host segments (Hastie, Tibshirani and Friedman, 2009). Figure 4 illustrates structural differences in listing supply across boroughs and accommodation types, revealing the uneven distribution of listings across the city. DBSCAN was also tested to evaluate density-based structure and identify potential outliers.

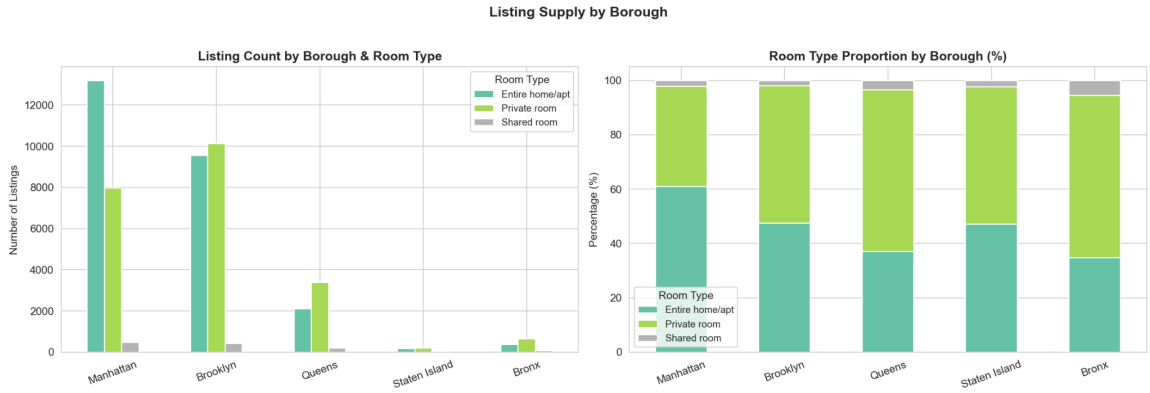

Figure 4. Supply distribution by borough and room type highlights structural market differences.

The optimal number of clusters was determined using the Elbow Method, which showed a clear inflection point at k = 4, indicating that four clusters provide the most meaningful balance between model complexity and explanatory power.

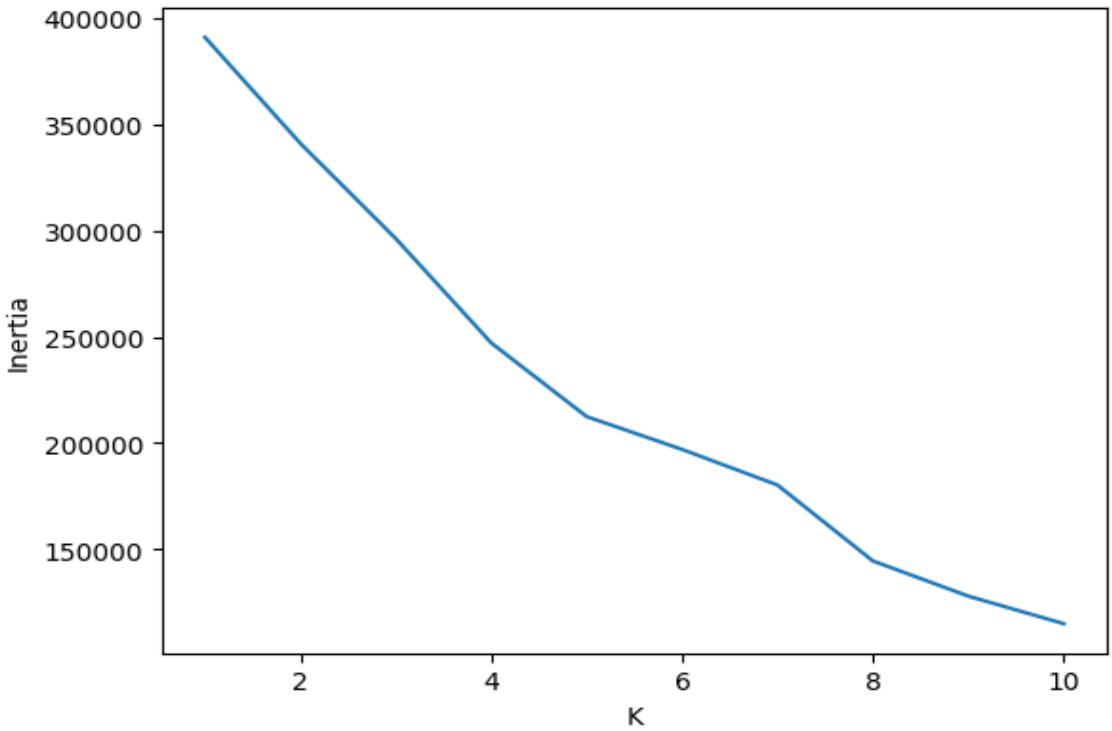

Figure 5. Elbow Plot for K-Means Cluster Selection.

Regression Analysis

Regression modelling is commonly used in Airbnb pricing studies to estimate how listing characteristics influence nightly rates (Perez-Sanchez, 2018). The multivariate linear regression model achieved an R^2 of approximately 0.53 with RMSE of around 0.475 on the log scale. This indicates moderate predictive performance and confirms that Airbnb pricing follows structured patterns while still being influenced by local variation, as illustrated by the predicted versus actual price relationship shown in Figure 6. Borough and room type were the strongest predictors of price, with Manhattan consistently commanding the highest prices, which is consistent with previous research showing that location and property characteristics are key determinants of Airbnb pricing (Gyódi and Nawaro, 2021). Spatial coordinates captured micro-location effects, reinforcing the importance of neighbourhood context.

Figure 6. Predicted vs actual log prices from the linear regression model.

Review activity showed a slight negative relationship with price, suggesting that lower priced listings may compete on occupancy and booking volume. Review indicators are often interpreted as demand and reputation signals that influence Airbnb pricing behaviour (Almeida, Nunes and Machado, 2025). Hosts managing multiple listings tended to charge higher prices, which is consistent with more commercial pricing behaviour.

Clustering Analysis

K-Means clustering identified four practical market segments. Low Activity Listings show weak demand signals and high availability, indicating underperformance. Standard Listings represent the core of the market with moderate prices and balanced activity. Investment Listings are the most commercial segment with high prices and large host

portfolios. High Demand Value Listings combine competitive prices with strong review activity, indicating high occupancy.

DBSCAN produced a low silhouette score of around 0.20, suggesting that the Airbnb market behaves more like a continuous economic spectrum than sharply separated natural clusters. K-Means therefore provided the most useful segmentation approach for management insights.

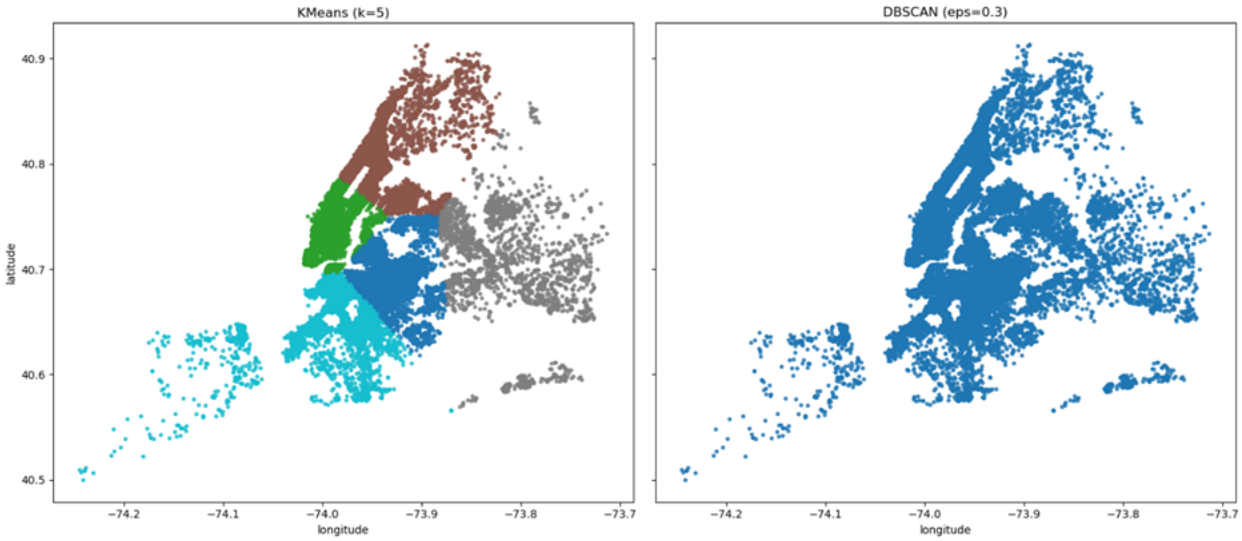

Figure 7. Comparison of spatial clustering results using K-Means and DBSCAN for Airbnb listings in New York City.

Business Insights

Airbnb should adopt a segment-aware pricing strategy rather than a single city-wide rule. Pricing guidance should prioritise borough, room type and micro-location because these variables carry the strongest pricing signal. High demand listings should focus on occupancy optimisation, while investment listings require yield and stay-length optimisation. Low activity listings should receive prompts encouraging improved listing quality and promotional pricing.

Strategic Insight

The analysis demonstrates that Airbnb pricing is primarily driven by structural market factors rather than simple numerical indicators. Location and accommodation type

explain much of the price variation, while behavioural indicators such as review frequency reflect demand and listing popularity.

Analytical Reflection

While the regression model explains a substantial proportion of price variation, residual variance suggests that additional contextual factors influence pricing. Property quality, amenities, host reputation and neighbourhood micro-features such as proximity to attractions or transport infrastructure likely contribute to pricing differences as these factors have been widely identified as important determinants of Airbnb pricing (Chang and Li, 2020). Future models could incorporate richer spatial variables or non-linear approaches such as gradient boosting to capture complex interactions between features (James et al., 2021).

Conclusion

This analysis shows that classical machine learning techniques can provide interpretable price prediction and meaningful market segmentation for Airbnb listings in New York City. Regression results indicate that pricing is primarily driven by structural factors such as borough and room type, while clustering reveals distinct listing segments reflecting different host strategies and demand patterns. These findings suggest that Airbnb should adopt a segment-aware and location-sensitive pricing strategy rather than relying on a single city-wide model.

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